Magnetostrictive transformer with two ultrasonic transducers

Does a magnetostrictive core carrying ultrasound mix it with the electrical drive, and does the swap test show it? — register entry H37

kronspice simulation report · 2026-11-07

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Summary

A transformer is wound on a magnetostrictive core — Terfenol-D or Galfenol, shaped as a rod, a ring or a helical spring — on a DC bias, and two ultrasonic transducers set up standing waves in the core. The primary is driven at 5 kHz (f_e) and the transducers at the core's modes (f_a, f_b). The question (H37, proposed after Barrett's H36) is whether the core mixes the sound with the electrical drive, and whether driving the secondary instead (the swap test) shows the nonlinearity. The standard prediction is computed with an energy-consistent magnetoelastic model.

- **Drive × sound products appear on the secondary:** $f_a \pm f_e$ and $f_b \pm f_e$ up to 2.9 mA (Terfenol-D ring, $f_b + f_e$), and only with both drive and sound on.
- **Sound × sound products are about 100× weaker** ($f_a + f_b$, $f_b - f_a$: 0.017 mA at most).
- **Reciprocal at small drive:** the transformer is reciprocal to 2.1×10^{-10} or better, the check that the magnetoelastic coupling is energy-consistent.
- **Not reciprocal at full drive for the products:** swapped, the Galfenol ring's summed products fall from 3.04 to 1.71 mA (×0.56) and the rod's from 0.388 to 0.290 mA (×0.75), while the transformer line stays nearly unchanged.
- **The Galfenol helix reads the ultrasound best** (20.0 mA at f_a); the Terfenol-composite helix is weakest (H/a = 0.02). The spring shape does not amplify.
- **Ledger residuals** 8×10^{-5} to 1×10^{-2} (first-order time stepping).

What is being tested

Hypothesis H37 (proposed 2026, after Barrett H36): a Terfenol-D or Galfenol core (rod, ring, helical spring) carrying standing waves from transducers at its ends mixes them with the electrical drive, and the swap test (drive the secondary, watch the primary) shows the nonlinearity. It is a mechanical relative of Barrett's oscillator-shuttle mixer (examples/barrett_osc).

Standard prediction (computed here): an anhysteretic magnetoelastic material mixes drive and sound through the curvature of its magnetization and strain curves. Products fall to zero at zero bias, where $\lambda' = 0$, and vanish when either the drive or the sound is off. The linear transfer must be reciprocal; the products need not be.

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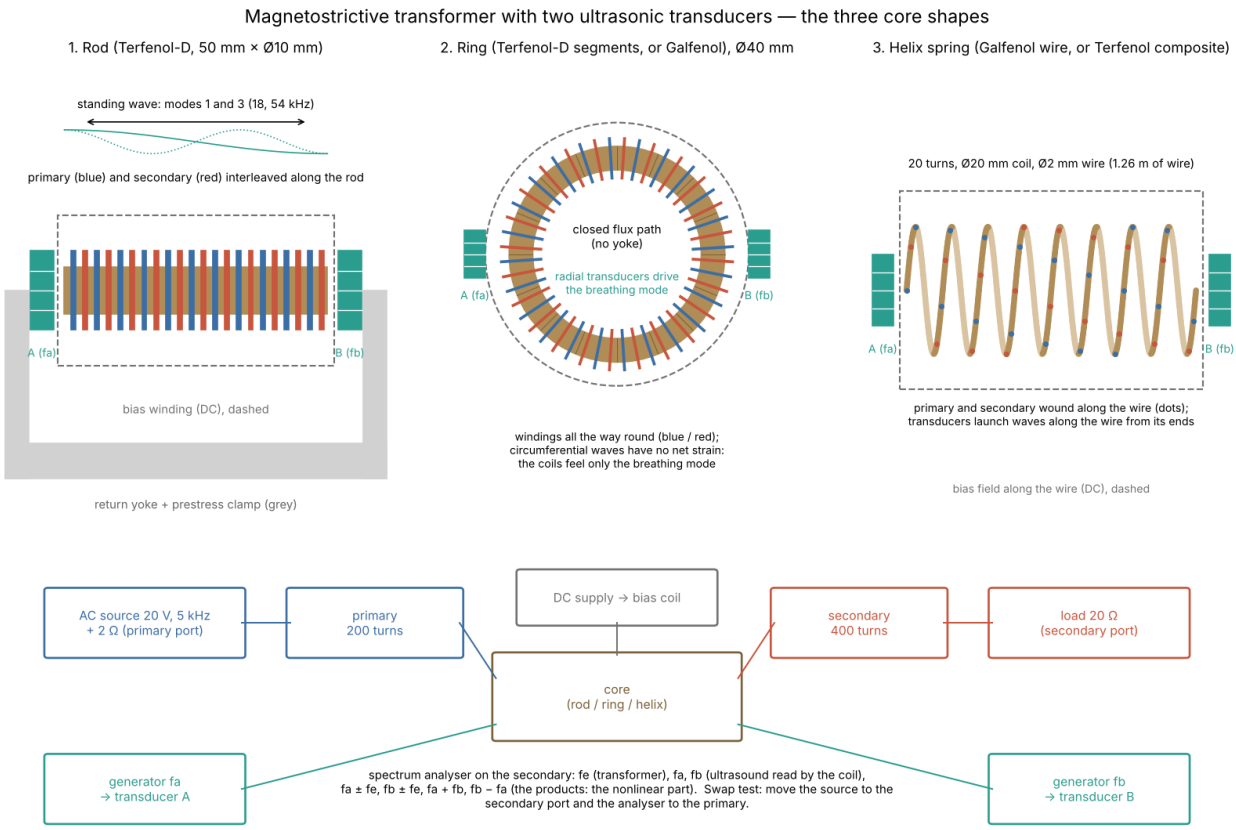


Figure 1. The set-up. Top: the three core shapes — (1) Terfenol-D rod, 50 mm × Ø10 mm, with primary (blue) and secondary (red) interleaved, a return yoke and prestress clamp, and transducers A and B at the ends (modes 1 and 3, 18 and 54 kHz); (2) ring, Ø40 mm, wound all the way round, radial transducers driving the breathing mode; (3) helix spring, 20 turns, Ø20 mm coil of Ø2 mm wire (1.26 m), windings along the wire. The DC bias winding is dashed. Bottom: the circuit — 20 V, 5 kHz source with 2 Ω on the 200-turn primary, 400-turn secondary into the 20 Ω load, DC bias coil, and two generators driving the transducers; the spectrum analyser watches the secondary.

The model

Material

An anhysteretic magnetization with quadratic magnetostriction, derived from one potential:

```
M = Ms·tanh(Ht/a),    λ = λm·tanh²(Ht/a),    Ht = H + Hbias
σ = E·(ε − λ),        B = μ0·(Ht + M) + σ·λ′(Ht)
Φ(Ht, ε) = ½E(ε−λ)² − μ0·(Ht²/2 + Ms·a·ln cosh(Ht/a)),    σ = ∂Φ/∂ε,  B = −∂Φ/∂Ht
```

Here M is the magnetization, M_s its saturation value, a the field scale of the curve, λ the magnetostrictive strain, σ the stress, ε the strain and E Young's modulus. Because both laws come from one potential, the magnetic and mechanical sides exchange energy exactly; the stored energy is Φ + H_t·B. Hysteresis and eddy currents are not modelled. The bias puts every material at H_b = 0.66·a, where λ′ = dλ/dH is largest.

material (declared)	d = λ′ at bias	μ_r	k	E	c	Q	bias	sources
Terfenol-D	13.3 nm/A	7.1	0.77	30 GPa	1801 m/s	10	57 kA/m	Etrema: μ_r 3–10, E 25–35 GPa, 800–1200 ppm
Terfenol-D/epoxy composite (51 %)	5.4 nm/A	3.7	0.31	15 GPa	1698 m/s	15	66 kA/m	measured d_{33} ≈ 5.5 nm/A, k_{33} ≈ 0.34
Galfenol ($Fe_{81}Ga_{19}$)	20.9 nm/A	87.1	0.49	60 GPa	2774 m/s	100	6 kA/m	Etrema: d_{33} 15–30 nm/A, μ_r 75–100, B_s 1.5 T, 7800 kg/m³

k is the magnetomechanical coupling factor, c the sound speed, Q the mechanical quality factor at the first tone (Kelvin–Voigt damping).

Cores, windings and numerics

- **Rod:** Terfenol-D, 50 mm × Ø10 mm; flux closes through a return yoke; a prestress clamp holds it. Modelled as a free–free line of segments, transducers at the two ends.

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- **Ring:** Ø40 mm, 5 × 5 mm; Terfenol as bonded segments or machined Galfenol. Modelled by its breathing mode, driven by radial transducers.
- **Helix:** 20 turns, Ø20 mm, of Ø2 mm wire (1.26 m), Galfenol or moulded Terfenol composite; a free–free line along the wire.
- **Windings:** uniform, so H is uniform along the core and the coils link its mean flux. Primary 200 turns, secondary 400 turns, each with 5 % leakage; ports 2 Ω (primary) and 20 Ω (load).
- **Drive:** 20 V at 5 kHz; transducers 5 N each. Tones: the rod's modes 1 and 3; the ring's breathing mode and 1.37× it; the helix wire's odd modes near 20 and 40 kHz. Even free–free modes and a ring's circumferential modes have no net strain, so uniform coils cannot feel them.
- **Numerics:** coil currents are stiff (leakage time constants can be nanoseconds), so they step by backward Euler and the mechanics by symplectic Euler. With a 4× finer step the lines move by 1–2 % and the ledger residual falls 4×, as a first-order scheme should.

Results

Forward run: drive the primary, watch the secondary

All lines are load current in mA (from run_output.txt). H/a is the peak field swing relative to the curve's field scale.

core	tones f_a, f_b [kHz]	H peak/a	f_e	f_a	f_b	ledger
Terfenol-D rod	18.0, 54.0	0.07	1397.810	3.1619	1.7808	1e-03
Terfenol-D ring	14.3, 19.6	0.11	1199.246	6.9186	13.6145	1e-02
Galfenol ring	22.1, 30.2	0.18	1334.504	3.4827	1.9444	5e-03
Terfenol-D composite helix	20.9, 39.9	0.02	13.139	0.7920	0.7493	8e-05
Galfenol helix	21.0, 40.8	0.17	271.884	20.0115	4.3118	2e-03

The mixing products on the same runs (mA):

core	f_a+f_e	f_a-f_e	f_b+f_e	f_b-f_e	f_a+f_b	f_b-f_a
Terfenol-D rod	2.2e-01	3.4e-02	7.8e-02	5.7e-02	2.1e-03	1.4e-03
Terfenol-D ring	7.7e-01	2.1e-01	2.9e+00	1.2e-01	1.7e-02	5.1e-03
Galfenol ring	5.6e-01	1.2e-01	1.5e-01	2.2e+00	4.1e-03	2.2e-03
Terfenol-D composite helix	2.8e-04	1.7e-04	4.6e-04	3.6e-04	3.4e-07	1.1e-07
Galfenol helix	1.7e+00	1.3e+00	4.4e-01	4.1e-01	5.3e-03	3.0e-03

What the secondary sees

- **Transformer line (f_e):** 1.2–1.4 A for the rod and rings. The helix's long, thin magnetic path gives a far smaller inductance and field (13.139 mA composite, 271.884 mA Galfenol).
- **Ultrasound itself (f_a, f_b):** the coil reads it through the Villari effect, 1–20 mA.
- **Products of drive and sound ($f_a \pm f_e$, $f_b \pm f_e$):** the nonlinear part, up to 2.9 mA. They appear only with both drive and sound on (TestRodAndRing), and are largest where the field swings furthest on the curve (H/a) and the coupling is strong.
- **Sound × sound products:** about 100× weaker. In this model the magnetization is linear in strain, so these products need the magnetic field as a go-between.
- **Rings** (closed flux path, no yoke) give the strongest products per amp. The Terfenol ring's f_b line (13.6 mA, off resonance) beats its f_a line (6.9 mA, at the breathing frequency): with the coils loaded the ring's resonance moves up by the ΔE effect, $1/\sqrt{(1 - k^2)} = 1.57$ for $k = 0.77$.
- **The spring shape does not amplify.** Uniform strain along the wire changes the coil's diameter, not its length. A spring's amplified motion comes from twisting the wire (the Wiedemann effect), which needs a current through the helix itself; that is not modelled here and is the next step.
- **Ring standing waves:** on a ring wound all the way round, circumferential standing waves cannot reach the coils; only the breathing mode couples. Sector windings would change that.

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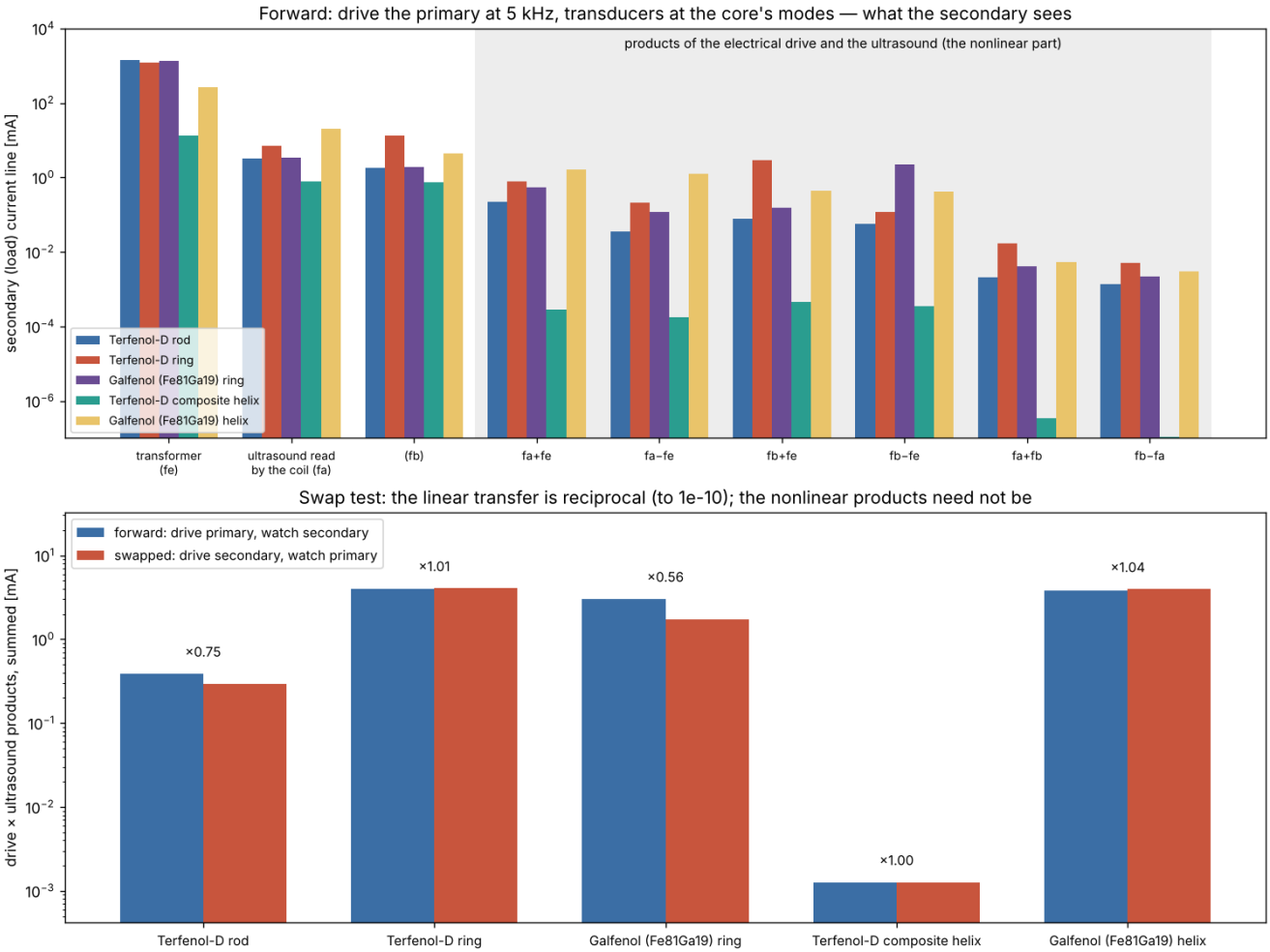


Figure 2. Top: forward run, secondary (load) current per line on a log scale for the five cores — the transformer line f_e, the ultrasound read by the coil (f_a, f_b) and, in the shaded band, the products of drive and sound and of sound with sound. Bottom: swap test, drive × ultrasound products summed, forward (blue) and swapped (red), with the swapped/forward ratio above each pair.

Swap test: drive the secondary, watch the primary

The same source is moved to the secondary port, in series with its 20 Ω. Reciprocity is checked at small drive; the line currents are at full drive.

core	reciprocity (small drive)	f_e fwd → rev [mA]	products $f_a \pm f_e$, $f_b \pm f_e$ summed: fwd → rev [mA]	ratio
Terfenol-D rod	3.8e-12	1397.810 → 1397.814	3.88e-01 → 2.90e-01	×0.75
Terfenol-D ring	4.8e-12	1199.246 → 1199.771	4.01e+00 → 4.06e+00	×1.01
Galfenol ring	1.2e-13	1334.504 → 1334.521	3.04e+00 → 1.71e+00	×0.56
Terfenol-D composite helix	2.0e-11	13.139 → 13.138	1.27e-03 → 1.27e-03	×1.00
Galfenol helix	2.1e-10	271.884 → 268.662	3.79e+00 → 3.95e+00	×1.04

Ratios as printed in the README and on Figure 2.

At small drive the transformer is reciprocal to 1e-10 or better. At full drive the transformer line stays almost reciprocal, but the products need not be: the Galfenol ring gives 44 % less going backwards, the rod 25 % less. The two directions put the core at different points of its curve (the windings' current, and so the field swing, differ) — a clean sign of the nonlinearity, which is what the swap test looks for.

Ledger and validation

- Energy ledger residuals per core: rod 1e-3, Terfenol ring 1e-2, Galfenol ring 5e-3, composite helix 8e-5, Galfenol helix 2e-3 (first-order scheme; the residual falls 4× with a 4× finer step).
- Lines converged to 1–2 % under a 4× finer step.
- Small-signal reciprocity 1.2e-13 to 2.1e-10: the magnetoelastic coupling is energy-consistent.

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- TestRodAndRing: the drive × sound products appear only with both the drive and the sound on.

Limits of the model

- Anhysteretic: no hysteresis loss and no eddy currents. Eddy currents are why the helix uses Terfenol composite; a solid Terfenol core would need laminating.
- Uniform windings only, so only modes with net strain couple; sector windings are not modelled.
- No Wiedemann (twist) effect: a current through the helix wire itself is not modelled.
- Material values are declared from data sheets and the cited measurement, not fitted to a bench core.
- First-order time stepping (ledger 1e-4 to 1e-2).

Bench notes

- **Mounting:** Terfenol-D is brittle (28 MPa in tension). Pre-compress it (a few to 10 MPa) and bond the transducers under that clamp.
- **Galfenol** can be machined into a ring or bent into a spring.
- **Bias:** set it with a DC coil or permanent magnets. The products vanish at zero bias, where $\lambda' = 0$.
- **Spectrum analyser:** watch $f_a \pm f_e$ and $f_b \pm f_e$ on the secondary. Repeat with the transducers off (products gone) and with the drive off (only f_a, f_b).

Status

H37: bounded. The standard prediction is computed: drive × sound products of up to 2.9 mA on the secondary, absent without sound; sound × sound 100× weaker; reciprocal at small drive, products not reciprocal at full drive (Galfenol ring ×0.56, rod ×0.75 when swapped). The spring shape does not amplify; amplification would need the Wiedemann twist with current through the helix, not modelled. A bench build compares its spectrum against these lines.

Reproducing the results

```
go run ./examples/magnetostrictive_transformer            # about 2 s
go run ./examples/magnetostrictive_transformer -data DIR  # also the figure data
python3 examples/magnetostrictive_transformer/viz/plot_lines.py DIR viz/lines.png
python3 examples/magnetostrictive_transformer/viz/setup.py viz/setup.png
go test ./examples/magnetostrictive_transformer
```


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